# Yoel Kim

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## Research Interests

- Software Engineering → Formal software verification; Software reverse engineering; Embedded software;
- Programming Languages → Static and dynamic program analysis; Programming-by-example.
- Theory of Computation → Automata learning; Abstraction.

## Education

*Ph.D. in Computer Science and Engineering, Kyungpook National University* *Mar 2023 – Feb 2027 (Expected)*
- Advisor: Yunja Choi
- Dissertation: Data-Driven Abstraction Techniques for Embedded Software Verification

*M.S. in Computer Science and Engineering, Kyungpook National University* *Mar 2021 – Feb 2023*
- Advisor: Yunja Choi
- Thesis: An Automated Stub Generation Approach Using Program Synthesis for Software Verification

*B.S. in Computer Science and Engineering, Kyungpook National University* *Mar 2017 – Feb 2021*
- Track: Global Software Convergence
- GPA: 3.95/4.3 (Summa Cum Laude)

## Ongoing Work

1. AutoCL: A Tool for Active Learning of Symbolic Automata from C Programs
   Yoel Kim and Yunja Choi
   (In preparation) **TACAS'27**: *International Conference on Tools and Algorithms for the Construction and Analysis of Systems* (Regular Tool Papers Track)
2. Learning Functional Specifications from Stateful Control Programs through Active Model Learning
   Yunja Choi and Yoel Kim
   (In preparation) *IEEE Access*, 2026
3. LLM-Based State Machine Generation Technique for Reactive Systems and Its Performance Evaluation (Extended version of the KCSE'26 paper)
   Seungbin Choi, Yoel Kim, and Yunja Choi
   (Submitted) *KIPS Transactions on Software and Data Engineering* (KTSDE), 2026

## International Conferences

1. Active Learning of Symbolic Automata for Reactive Programs via Dynamic Symbolic Mapper
   Yoel Kim and Yunja Choi
   **FSE'26**: *ACM International Conference on the Foundations of Software Engineering*
2. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software
   Yoel Kim and Yunja Choi
   **FSE'24**: *ACM International Conference on the Foundations of Software Engineering*

## Domestic Conferences

1. LLM-Based State Machine Generation Technique for Reactive Systems and Its Performance Evaluation
   Seungbin Choi, Yoel Kim, and Yunja Choi
   KCSE'26: *Korea Conference on Software Engineering* ** Short Paper Award*
2. An Approach of Incremental Constraint Extraction Based on I/O Examples for Automatic Stub Generation
   Yoel Kim and Yunja Choi
   KCSE'23: *Korea Conference on Software Engineering* ** Best Short Paper Award*

3. A Case Study to Improve the Efficiency of Model Checking in Embedded Software Using Program Synthesis
   Yoel Kim and Yunja Choi
   KSC'21: *Korea Software Congress*

4. A Case Study on the Performance of Program Synthesis in Embedded Software Domain
   Yoel Kim and Yunja Choi
   KCSE'21: *Korea Conference on Software Engineering*

## Talks

1. AutoCL: A Tool for Active Learning of Symbolic Automata from C Programs. 11th STAAR Workshop. *Kyungpook National University, Daegu, Korea. Jul 14, 2026.*
2. Active Learning of Symbolic Automata for Reactive Programs via Dynamic Symbolic Mapper. Research paper presentation. FSE'26. *Concordia University, Montreal, Canada. Jul 7, 2026.*
3. AutoCL: 동적 심볼릭 매퍼 기반 심볼릭 오토마타 능동학습. 10th STAAR Workshop. *Ulsan, Korea. Feb 3, 2026.*
4. 동적 심볼릭 매퍼 기반 심볼릭 오토마타 능동학습. PhD 발표 세션. BK21 참여대학원생 컴퓨터학부 교류 학술대회. *Daegu, Korea. Jan 15, 2026.*
5. C2FSM: A Tool for Active Learning of Extended Finite State Machines from C Programs. 9th STAAR Workshop. *Korea University, Seoul, Korea. Jul 29, 2025.*
6. C2FSM: A Tool for Active Learning of Extended Finite State Machines from C Programs. PhD 발표 세션. BK21 참여대학원생 컴퓨터학부 교류 학술대회. *Daegu, Korea. Jul 21, 2025.*
7. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Top Conference Session. KCC'25. *Jeju, Korea. Jul 4, 2025.*
8. 프로그램 합성 기법을 이용한 효율적인 능동 모델 학습 방법. Lightning talk. 8th STAAR Workshop. *Yeosu, Korea. Feb 7, 2025.*
9. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Top Conference Session. KCSE'25. *Pyeongchang, Korea. Jan 22, 2025.*
10. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. 우수 논문 발표 세션. BK21 참여대학원생 컴퓨터학부 교류 학술대회. *Daegu, Korea. Jan 21, 2025.*
11. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Research paper presentation. FSE'24. *Porto de Galinhas, Ipojuca, Pernambuco, Brazil. Jul 17, 2024.*
12. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Published Paper Session. 7th STAAR Workshop. *Gyeongju, Korea. Jul 9, 2024.*
13. 능동 학습 기반 SW 재난 모델 자동생성. Lightning talk. 6th STAAR Workshop. *KAIST, Daejeon, Korea. Jan 30, 2024.*
14. 효율적인 검증 속성 위반 탐지를 위한 PBE 기반 함수요약 및 정제 기법. Lightning talk. 5th STAAR Workshop. *Ulsan, Korea. Jul 5, 2023.*
15. An Approach of Incremental Constraint Extraction Based on I/O Examples for Automatic Stub Generation. Research paper presentation. KCSE'23. *Pyeongchang, Korea. Feb 10, 2023.*
16. PBE 기반 스텁 자동생성 및 스텁코드 분할을 통한 모델검증 효율 향상. Lightning talk. 4th STAAR Workshop. *Paju, Korea. Feb 1, 2023.*
17. 프로그램 합성 기법을 이용한 모델체킹 검증 효율 향상 및 반례 트레이스 기반 알람 필터링 기법 연구. Lightning talk. 3rd STAAR Workshop. *Pohang, Korea. Jul 14, 2022.*
18. A Case Study to Improve the Efficiency of Model Checking in Embedded Software Using Program Synthesis. Research paper presentation. KSC'21. *Pyeongchang, Korea. Dec 21, 2021.*
19. A Case Study on the Performance of Program Synthesis in Embedded Software Domain. Research paper presentation. KCSE'21. *Online. Feb 2, 2021.*

## Experience

**Teaching Assistant**, Kyungpook National University *Mar 2021 – Jun 2024*

- ITEC0414: Software Testing Theory (Spring 2022, 2023, 2024).
- COMP0224: Software Design (Fall 2021, 2022).
- COMP0216: Data Structure Applications (Spring 2021).

**Undergraduate Tutor**, Kyungpook National University *Mar 2018 – Dec 2020*

- COMP0322: Database Management Systems (Fall 2020).
- CLTR266: Software Computational Thinking (Spring 2018, 2019).
- CLTR268: Python Programming (Winter 2018).
- COMP0216: Data Structure Applications (Fall 2018).

**Undergraduate Research Intern**, Software Safety Engineering Lab *Jul 2019 – Dec 2019; Sep 2020 – Feb 2021*

- Supervisor: Yunja Choi
- Developed a Java-based GUI tool to support software testing activities for Hyundai Motor.

**Intern**, JS System, Daegu *Dec 2019 – Feb 2020*

- Assisted with testing an FPGA-based embedded board.

**Undergraduate Research Intern**, AI-Based Networks Lab *Oct 2018 – Dec 2018*

- Supervisor: Dongkyun Kim